

Natural densities of the three additive complexities of the Tribonacci word

A proposed resolution of the proportion question
in Popoli–Shallit–Stipulanti, Remark 19

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Abstract

The Tribonacci word is the fixed point of $0 \mapsto 01$, $1 \mapsto 02$, $2 \mapsto 0$. Its additive complexity $a(n)$ counts the distinct sums of length- n factors. Popoli, Shallit and Stipulanti proved that the only values for positive n are 3, 4, 5, and asked for their respective proportions. We give an exact finite-certificate proof of the existence of the ordinary natural densities and evaluate them in the cubic field of the Tribonacci constant β , where $\beta^3 = \beta^2 + \beta + 1$:

$$d_3 = \frac{21\beta^2 - 38\beta + 5}{22}, \quad d_4 = \frac{-55\beta^2 + 72\beta + 67}{22}, \quad d_5 = \frac{17\beta^2 - 17\beta - 25}{11}.$$

For every integer cutoff N , not merely for Tribonacci-number cutoffs, the count of lengths with value j is $d_j N + O(N^\theta)$, where $\theta = \log(7/5)/\log \beta < 0.553$. The proof reconstructs a 76-state output automaton using abelian co-decomposition, verifies an exact algebraic left eigenvector and an integer matrix annihilator, and controls arbitrary cutoffs by a disjoint prefix-cylinder decomposition. The package includes the full transition and eigenvector tables, a standard-library-only verifier, exact counting code, and a separate all-factor check through length 2048.

Status. This is a proposed solution with reproduced exact finite certificates. It has not been independently refereed or formalized in a proof assistant. The literature audit found the explicit open question and no later resolution in the sources checked; it is not a guarantee of global priority. The known range $\{3, 4, 5\}$ and the co-decomposition method are not claimed as new.

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1 The problem and the result

1.1 The invariant being counted

Let φ be the substitution

$$\varphi(0) = 01, \quad \varphi(1) = 02, \quad \varphi(2) = 0, \quad (1)$$

and let

$$\mathbf{t} = \lim_{k \rightarrow \infty} \varphi^k(0) = 010201001020101020100 \dots$$

be its one-sided fixed point. Positions are indexed from zero. We write $\mathbf{t}[i : i + n]$ for the factor starting at i and having length n . For a finite word $u = u_0 \dots u_{m-1}$ on $\{0, 1, 2\}$, set

$$\sigma(u) = \sum_{i=0}^{m-1} u_i, \quad \Psi(u) = (|u|_0, |u|_1, |u|_2).$$

The additive complexity is

$$a(n) = \#\{\sigma(\mathbf{t}[i : i + n]) : i \geq 0\}. \quad (2)$$

Thus $a(n)$ counts *different sums*, not factors, occurrences, or Parikh vectors. In particular, two different Parikh vectors can contribute the same additive class. We use $a(0) = 1$ for the empty factor.

For example, the letters all occur, so $a(1) = 3$. The length-two factors are 00, 01, 02, 10, 20; their sums are 0, 1, 2, so $a(2) = 3$. By contrast, there are five distinct length-two factors and three distinct length-two Parikh vectors. These distinctions matter when interpreting a frequency question.

For $j \in \{3, 4, 5\}$ define

$$C_j(N) = \#\{0 \leq n < N : a(n) = j\}. \quad (3)$$

The variable being sampled uniformly is the *length* n , not the starting position of a factor. We seek the limits $C_j(N)/N$.

1.2 The published question and the exclusion check

In Theorem 18 of their full preprint [1], Popoli, Shallit and Stipulanti establish the range $\{3, 4, 5\}$ for positive lengths and provide a 76-state Tribonacci automaton computing the function. Their Remark 19 then leaves the respective proportions open. Our question is precisely that proportion question, with “proportion” made explicit as ordinary natural density over integer lengths.

The area was selected before this question was chosen: a single call to Python’s OS-backed `secrets.randbelow(120)` returned zero-based index 67, selecting *Combinatorics on words*, item 68 of the recorded 120-area list. The complete list and result are in `data/selection.json`. The uploaded catalogue `manifest(1).tex`, containing 71 research packages, was read in full. This target is not among its entries. Its transfinite-word order-type problems and its discrepancy problem for $0 \mapsto 1, 1 \mapsto (10)^m$ concern different substitutions, invariants and questions.

A focused current-literature search on 20 September 2026 found no later resolution of this particular density problem. The checked sources include the full 2024 preprint and the later work of Coudreau et al. [3], in its April 2025 version 2. The latter develops more general automata constructions for abelian-type complexities; it is not used as a proof of the densities below. The scope and exact source versions are recorded in `notes/sources.md`. Failure to find another solution is not itself evidence sufficient to assert absolute novelty.

1.3 Main theorem

Let β be the real root greater than one of

$$f(x) = x^3 - x^2 - x - 1. \quad (4)$$

Numerically, $\beta = 1.83928675521416113255\dots$

Theorem 1.1 (Natural densities and a power-saving error). *For the additive complexity (2), the ordinary natural densities of the values 3, 4, 5 exist and equal*

$$d_3 = \frac{21\beta^2 - 38\beta + 5}{22}, \quad (5)$$

$$d_4 = \frac{-55\beta^2 + 72\beta + 67}{22}, \quad (6)$$

$$d_5 = \frac{17\beta^2 - 17\beta - 25}{11}. \quad (7)$$

More precisely, with

$$\theta = \frac{\log(7/5)}{\log \beta} = 0.55215697321855134306\dots, \quad (8)$$

we have, for each $j \in \{3, 4, 5\}$,

$$C_j(N) = d_j N + O(N^\theta) \quad (N \rightarrow \infty). \quad (9)$$

All three densities are strictly positive and their sum is one.

The decimal values are

Value	Natural density	Percentage
3	0.27952701944967565180...	27.9527019449...%
4	0.60749905184438815304...	60.7499051844...%
5	0.11297392870593619515...	11.2973928705...%

The main point beyond extracting a dominant eigenvector is that (9) holds at *every* cutoff. Limits along a sparse sequence of numeration-system cutoffs do not by themselves establish natural density. Section 5 supplies the missing uniform step.

1.4 What is new, what is reused, and what is certified

The range of $a(n)$ and the existence of a 76-state automaton are already known [1]. Abelian co-decomposition is a known mechanism, developed for this setting by Turek [2]. We specialize and prove the needed recursions, then reconstruct the additive-complexity machine independently. We do not claim that our state numbering or serialized machine is identical to the authors' original file.

The proposed contribution is the exact evaluation of the three densities, the uniform all-cutoff proof, and the explicit power-saving estimate. The finite parts of the argument are exposed as reproducible certificates: a closed family of co-decompositions, the quotient map to the displayed machine, an algebraic eigenvector, and a polynomial identity for the integer adjacency matrix. A verifier checks those finite identities using only Python's standard library. The passage from them to an infinite-word theorem and a limit theorem is proved in the text.

2 Tribonacci numeration and word prefixes

Define words and lengths

$$W_k = \varphi^k(0), \quad T_k = |W_k|.$$

The initial values and recursions are

$$T_0 = 1, \quad T_1 = 2, \quad T_2 = 4, \quad T_k = T_{k-1} + T_{k-2} + T_{k-3} \quad (k \geq 3), \quad (10)$$

and

$$W_k = W_{k-1}W_{k-2}W_{k-3} \quad (k \geq 3). \quad (11)$$

Indeed, $\varphi^k(0) = \varphi^{k-1}(0)\varphi^{k-1}(1)$, $\varphi^{k-1}(1) = \varphi^{k-2}(0)\varphi^{k-2}(2)$, and $\varphi^{k-2}(2) = \varphi^{k-3}(0)$.

Lemma 2.1 (Canonical representations). *For every $k \geq 0$, the binary strings of length k with no occurrence of 111 represent the integers $0 \leq n < T_k$ exactly once under*

$$\text{val}(b_{k-1} \cdots b_0) = \sum_{i=0}^{k-1} b_i T_i.$$

Their lexicographic order agrees with numerical order. Leading zeros do not change the represented number. Every positive integer therefore has a unique representation beginning with 1 and avoiding 111.

Proof. The assertions are immediate for $k = 0, 1, 2$. For $k \geq 3$, the allowed strings split into the three disjoint prefix classes 0, 10, and 110. Their suffixes have lengths $k-1, k-2, k-3$, respectively, and may be arbitrary allowed strings of those lengths. By induction their represented intervals are

$$[0, T_{k-1}), \quad [T_{k-1}, T_{k-1} + T_{k-2}), \quad [T_{k-1} + T_{k-2}, T_k).$$

These intervals are adjacent, disjoint and ordered exactly as the three prefixes. This proves all assertions, including uniqueness and order. $\square$

Write $P_n = \mathbf{t}[0 : n]$ and use $\text{rep}(n)$ for the canonical representation, with $\text{rep}(0)$ the empty string.

Lemma 2.2 (The prefix associated with a representation). *If $\text{rep}(n) = b_s \cdots b_0$, then*

$$P_n = W_s^{b_s} W_{s-1}^{b_{s-1}} \cdots W_0^{b_0}. \quad (12)$$

Consequently, if appending $d \in \{0, 1\}$ gives another allowed representation, and $N = \text{val}(\text{rep}(n)d)$, then

$$P_N = \varphi(P_n)0^d. \quad (13)$$

Proof. The base cases follow from $W_0 = 0$, $W_1 = 01$, $W_2 = 0102$. For a prefix shorter than T_k , the same three intervals in the proof of Lemma 2.1 say that it lies in the first block of (11), consists of that block and a prefix of the second, or consists of the first two and a prefix of the third. Induction within those blocks gives (12). Applying φ replaces each W_i by W_{i+1} ; an appended digit d contributes $W_0^d = 0^d$. This proves (13). $\square$

The generating function of the numeration weights is

$$\sum_{k \geq 0} T_k z^k = \frac{1 + z + z^2}{1 - z - z^2 - z^3}. \quad (14)$$

For later use it is enough that $T_k \asymp \beta^k$. This also follows below from the exact automaton spectrum; no asymptotic estimate is being assumed without proof.

3 From all factors to a finite automaton

3.1 Co-decomposition and relative sums

Definition 3.1. For finite words u, v with $\Psi(u) = \Psi(v)$, cut both words at every position i at which $\Psi(u[0 : i]) = \Psi(v[0 : i])$. The set of paired nonempty pieces so obtained is denoted $\text{Dec}(u, v)$. Repeated identical pairs are retained only once. This is the finest, or maximal-cut, abelian co-decomposition.

The use of a set is intentional: only which relative vectors occur matters, not how many times a particular piece appears.

For a pair (u, v) of equal length define its scalar prefix differences by

$$S(u, v) = \{\sigma(v[0 : i]) - \sigma(u[0 : i]) : 0 \leq i \leq |u|\},$$

and for a finite set Z of such pairs let

$$S(Z) = \bigcup_{(u,v) \in Z} S(u, v), \quad o(Z) = |S(Z)|. \quad (15)$$

If a pair is cut at a common Parikh prefix, the relative vector, and hence the scalar difference, is zero at the cut. It follows that $S(u, v)$ is exactly the union of the scalar-difference sets of its pieces.

3.2 A finite word containing every relevant factor

Fix $n \geq 1$, choose K such that $n \leq T_K$, and put

$$W = \varphi^{K+3}(0), \quad \text{rot}_n(W) = P_n^{-1}WP_n.$$

Here $P_n^{-1}W$ means deletion of the specified prefix, not a group inverse. The prefix is available because P_n is a prefix of W .

Lemma 3.2 (Coverage and independence). *The set*

$$Z(n) = \text{Dec}(W, \text{rot}_n(W)) \quad (16)$$

is independent of the choice of K satisfying $n \leq T_K$. Moreover,

$$a(n) = o(Z(n)). \quad (17)$$

Proof. Set $B_a = \varphi^{K+1}(a)$ for $a \in \{0, 1, 2\}$. Each B_a begins with W_K : explicitly, $B_2 = W_K$, while B_0 and B_1 begin with W_K . Thus each starts with P_n and has length at least n . The fixed point is a concatenation of these blocks. A length- n factor starting inside B_a is contained in B_aP_n , because any part extending beyond the block is a prefix of the next block of length at most n .

Now $W = \varphi^{K+1}(0102)$ contains every block type B_a . Every length- n factor is therefore found at a position $0 \leq i < |W|$ in WP_n . Also WP_n really is a prefix of $\mathbf{t}$: the letter following the first 0 at the substitution level is 1, and $\varphi^{K+3}(1)$ begins with W_{K+2} , which is long enough to contain P_n .

For $0 \leq i \leq |W|$, subtraction of prefix vectors gives

$$\begin{aligned} \Psi(\text{rot}_n(W)[0 : i]) - \Psi(W[0 : i]) &= \Psi(\mathbf{t}[n : n + i]) - \Psi(\mathbf{t}[0 : i]) \\ &= \Psi(\mathbf{t}[i : i + n]) - \Psi(P_n). \end{aligned}$$

As i varies, these are exactly the relative Parikh vectors of length- n factors; the possible endpoint $i = |W|$ contributes the already present zero vector. Cutting at every zero vector does not change the union of the prefix-difference vectors. Applying the linear functional $(x_0, x_1, x_2) \mapsto x_1 + 2x_2$ and adding the constant $\sigma(P_n)$ therefore proves (17).

For independence, let K_0 be the least index with $n \leq T_{K_0}$ and instead fix $B_a = \varphi^{K_0+1}(a)$. For every $K \geq K_0$ we have

$$\varphi^{K+3}(0) = \varphi^{K_0+1}(\varphi^{K-K_0+2}(0)).$$

The word inside the parentheses contains all three letters, since $\varphi^2(0) = 0102$. Each B_a starts with P_n . If $B_a = P_n U_a$, then rotating a concatenation of these blocks by P_n gives the concatenation of the corresponding $U_a P_n$. At every block boundary the top and bottom have the same Parikh vector. The finest co-decomposition is consequently

$$\bigcup_{a \in \{0,1,2\}} \text{Dec}(B_a, P_n^{-1} B_a P_n),$$

which no longer depends on K . This proves independence. $\square$

3.3 The digit transition

The following specializes the co-decomposition mechanism of Turek [2]; we give the argument because it is the connection between the infinite word and the finite certificate.

For a pair (u, v) define

$$D_0(u, v) = \text{Dec}(\varphi(u), \varphi(v)), \quad (18)$$

$$D_1(u, v) = \text{Dec}(\varphi(u), 0^{-1}\varphi(v)0). \quad (19)$$

Every nonempty image under φ begins with 0, so the prefix deletion in (19) is always legal. Extend the operation to a set by $D_d(Z) = \bigcup_{(u,v) \in Z} D_d(u, v)$.

Proposition 3.3 (Exact finite-state update). *We have*

$$Z(1) = \{(0, 0), (01, 10), (02, 20)\}. \quad (20)$$

If $\text{rep}(N) = \text{rep}(n)d$ is allowed and $n \geq 1$, then

$$Z(N) = D_d(Z(n)). \quad (21)$$

Proof. The initial calculation uses $W = \varphi^3(0) = 0102010$ and its one-letter rotation 1020100 ; cutting at all equal Parikh prefixes gives (20).

Choose K sufficiently large that both the old and the new prefix fit in the block constructions of Lemma 3.2. By Lemma 2.2, $P_N = \varphi(P_n)0^d$. For $d = 0$, applying φ to the old paired words gives the new rotated pair. All former piece boundaries remain common Parikh boundaries, so refinement inside each image pair gives exactly $D_0(Z(n))$.

For $d = 1$, the new bottom word is the additional one-letter rotation of the bottom word just obtained. If its consecutive image pieces are $\varphi(v_1), \dots, \varphi(v_m)$, each starts with 0. Writing each as $0V_i$ shows

$$0^{-1}((0V_1) \cdots (0V_m))0 = (V_1 0) \cdots (V_m 0).$$

Thus the one-letter rotation can be performed in each piece separately. Each rotated piece still has the same Parikh vector as its top partner. Refinement at all common Parikh prefixes gives $D_1(Z(n))$, as required. $\square$

3.4 Exhaustive closure and the 76-state quotient

Starting with (20), apply both transitions to every newly encountered set until no new set remains. The executed construction reaches exactly 277 distinct sets, using 56 distinct word pairs; the longest word in a pair has length 24. These are a finite closed family, not a sample of the first few lengths. The serialized family, its two outgoing transitions at every state, and its scalar output sets are all in `data/co_decomposition.json`.

The raw closure is larger than necessary because it also allows digit strings containing 111. To restrict to the canonical language, form the product with the trailing-one counter $r \in \{0, 1, 2\}$. A zero resets r ; a one increments it, and a one is forbidden when $r = 2$. Beginning after the first digit 1 gives 296 reachable product states. Add a zero/empty initial state of output 1 and an invalid sink of output 0.

Partition states by output and repeatedly refine using the signature

$$(\text{output}(q), \text{class}(\delta(q, 0)), \text{class}(\delta(q, 1))).$$

After stabilization, the quotient has 77 states including the invalid sink. Omitting the sink leaves 76 live states. Canonical breadth-first numbering, visiting digit 0 before digit 1, gives the table in Appendix A. Its initial state is 0, and a dash denotes the omitted invalid transition.

Proposition 3.4 (Certified machine). *For every $n \geq 0$, reading $\text{rep}(n)$ from state 0 in the table of Appendix A produces the output $a(n)$. Leading zeros may be inserted arbitrarily. In particular $a(n) \in \{3, 4, 5\}$ for $n > 0$.*

Proof. The finite closure certificate checks (18)–(19) for all 277 sets. Proposition 3.3 and induction on the number of digits show that its output is $a(n)$ on every allowed positive representation. The language product forbids exactly 111. At every refinement step, states in one stable class have equal outputs and their transitions go to equal classes. The included quotient map checks this commuting property for *every* product state and both digits, not only for representatives. Thus quotienting preserves all outputs on all finite inputs. The added initial state handles the empty string and leading zeros. Finally, all positive-representation states have output 3, 4, or 5, as the table shows. $\square$

This proposition reproves the already known range theorem. It is included to make the spectral calculation independent of an unavailable external automaton file. The implementation is short enough to audit: it uses only word substitution, integer prefix differences, finite sets, a queue, and partition refinement. No search through an unbounded sequence is used to guess a transition.

4 An exact spectral certificate

4.1 The adjacency matrix and suffix counts

Let $Q = \{0, 1, \dots, 75\}$ be the live state set, let $o(q)$ denote its output, and define the nonnegative integer matrix

$$A_{qt} = \#\{d \in \{0, 1\} : \delta(q, d) = t\}. \quad (22)$$

An illegal transition contributes nothing. If both digits have the same target, their contributions are counted with multiplicity. Row vectors act on the left. For each $j \in \{1, 3, 4, 5\}$, let h_j be the column vector whose q th entry is 1 when $o(q) = j$ and 0 otherwise. Then

$$e_q A^k h_j \quad (23)$$

counts the legal suffixes of length k that, starting at q , finish at output j . In particular,

$$C_j(T_k) = e_0 A^k h_j, \quad T_k = e_0 A^k \mathbf{1} \quad (24)$$

for $j = 3, 4, 5$.

The key task is to understand (23) *uniformly in q* . There are only 76 states, but their leading amplitudes are not all the same. Ignoring that dependence would make the all-cutoff argument invalid.

4.2 The closed component and a right eigenvector

The transition table has the closed strongly connected component

$$\mathcal{C} = \{5, 9, 10, 11\} \cup \{14, 15, \dots, 75\}. \quad (25)$$

It contains 66 states, and every live state can reach it. State 62 has a self-loop on digit 0. These assertions are checked by forward and reverse graph searches in the verifier. Within $\mathcal{C}$, there are 10 states of output 3, 41 of output 4, and 15 of output 5. *The desired proportions are not these counts divided by 66.* Different states are reached with different asymptotic weights.

Each live state has a well-defined trailing-one context $r(q) \in \{0, 1, 2\}$. It can be recovered directly from the table: $r(q) = 2$ if its 1-transition is illegal; otherwise $r(q) = 1$ if the target's 1-transition is illegal; and otherwise $r(q) = 0$. The table verifies

$$r(\delta(q, 0)) = 0, \quad r(\delta(q, 1)) = r(q) + 1$$

whenever the latter transition is legal. Define

$$v_q = \begin{cases} 1, & r(q) = 0, \\ \beta - 1, & r(q) = 1, \\ \beta^{-1} = \beta^2 - \beta - 1, & r(q) = 2. \end{cases} \quad (26)$$

All entries are positive, and $v_0 = 1$.

Lemma 4.1. *The vector (26) satisfies $Av = \beta v$.*

Proof. At context 0, the two target values sum to $1 + (\beta - 1) = \beta$. At context 1, they sum to $1 + \beta^{-1} = \beta^2 - \beta = \beta(\beta - 1)$. At context 2, only the zero-transition remains, with value $1 = \beta\beta^{-1}$. These are exactly the three eigenvector equations. The identity $1 + \beta^{-1} = \beta^2 - \beta$ follows from (4). $\square$

4.3 A left eigenvector in the cubic field

Appendix B supplies integer triples (u_q, w_q, z_q) with

$$\ell_q = \frac{u_q + w_q\beta + z_q\beta^2}{44}. \quad (27)$$

The ten entries outside $\mathcal{C}$ are zero. The exact finite identities are

$$\ell A = \beta \ell, \quad \ell \mathbf{1} = 1. \quad (28)$$

No floating-point calculation is needed to verify them. Reduction modulo $f(\beta) = 0$ sends multiplication by β to the integer operation

$$(u, w, z) \mapsto (z, u + z, w + z). \quad (29)$$

For each column q of A , add the triples at its incoming sources, with edge multiplicities, and compare with (29). This is a collection of 228 integer component equalities. Normalization amounts to

$$\sum_q (u_q, w_q, z_q) = (44, 0, 0).$$

For a concrete example, the incoming sources for state 62 are 56, 62, 71. Its triple is $(14, -48, 22)$, and the sum of the three incoming triples is $(22, 36, -26)$, exactly its image under (29).

The sums over output classes are particularly simple:

Output	$\sum u_q$	$\sum w_q$	$\sum z_q$
1	0	0	0
3	10	-76	42
4	134	144	-110
5	-100	-68	68

Consequently

$$\ell h_j = d_j \quad (j = 3, 4, 5) \quad (30)$$

with the expressions in Theorem 1.1. Direct multiplication of the left and right vectors, reduced using (4), also gives

$$\ell v = 5\beta^2 - 6\beta - 5, \quad \frac{1}{\ell v} = \frac{3\beta^2 + 7\beta + 2}{22} > 0. \quad (31)$$

The second identity can be verified by multiplying its two sides by $5\beta^2 - 6\beta - 5$ and reducing modulo f .

4.4 An integer annihilating polynomial

The next certificate controls every eigenvalue, including transient states. Define

$$\begin{aligned} p(x) = & x^{45}(x-1)^3(x^2+x+1)(x^8-x^4+1) \\ & \cdot (x^3-x^2-x-1)(x^3+x^2+x-1) \\ & \cdot (x^{12}-2x^9-2x^5-2x+1). \end{aligned} \quad (32)$$

It has degree 76.

Lemma 4.2 (Finite annihilator certificate). *The matrix in (22) satisfies*

$$p(A) = 0. \quad (33)$$

Exact verification. Expand the product (32) by integer convolution, then evaluate it at A by Horner's rule. Each left multiplication by A is only the addition of the rows corresponding to the one or two legal targets in the transition table. Every one of the $76^2 = 5776$ final entries is zero. The routine `check_spectral` in `code/verify.py` performs these operations verbatim, with unbounded integers. Thus (33) is a finite explicit integer identity, not a numerical eigenvalue estimate. An optional symbolic derivation also identifies p as the characteristic polynomial, but that stronger identification is unnecessary for this proof. $\square$

Lemma 4.3 (Separation from the dominant root). *Every zero of p other than β has modulus strictly less than $7/5$. The zero β has multiplicity one in p .*

Proof. The zero factors at 0 and the roots of unity are harmless. The roots of $x^8 - x^4 + 1$ lie on the unit circle because

$$(x^4 + 1)(x^8 - x^4 + 1) = x^{12} + 1.$$

For $f(x) = x^3 - x^2 - x - 1$, its derivative is $(3x + 1)(x - 1)$, and its local maximum, at $-1/3$, is $-22/27 < 0$. Hence f has just one real root. Direct substitution places this root in $(9/5, 19/10)$. The other roots are a conjugate pair whose product is $1/\beta$, so each has modulus $\beta^{-1/2} < 1$. The roots of $x^3 + x^2 + x - 1$ are the reciprocals of the roots of f . Their moduli are β^{-1} and $\sqrt{\beta}$, both less than $7/5$, since $\beta < 19/10 < 49/25$.

It remains to treat

$$R(x) = x^{12} - 2x^9 - 2x^5 - 2x + 1.$$

At $r = 7/5$, the following strictly positive margin is exact:

$$r^{12} - 2r^9 - 2r^5 - 2r - 1 = \frac{199057326}{244140625} > 0. \quad (34)$$

If $R(z) = 0$ and $|z| \geq r$, the triangle inequality would give

$$1 \leq 2|z|^{-3} + 2|z|^{-7} + 2|z|^{-11} + |z|^{-12} \leq 2r^{-3} + 2r^{-7} + 2r^{-11} + r^{-12} < 1,$$

a contradiction. This proves the modulus bound for the last factor. All other factors are nonzero at β , and the real root of f is simple, proving the multiplicity assertion. $\square$

4.5 The rank-one asymptotic and uniform cancellation

Lemma 4.4 (Simple dominant eigenvalue). *The eigenspace of A at β is one-dimensional, spanned by v . Its algebraic multiplicity is also one.*

Proof. Given an eigenvector x , divide its coordinates by the positive vector v , writing $y_q = x_q/v_q$. The equation $Ax = \beta x$ becomes

$$y_q = \sum_t P_{qt} y_t, \quad P_{qt} = \frac{A_{qt} v_t}{\beta v_q}.$$

By Lemma 4.1, every row of P sums to one. On the closed strongly connected set $\mathcal{C}$, a real harmonic function is constant: at a maximum, every successor with positive transition probability must have that same value, and connectivity propagates it throughout $\mathcal{C}$. For a complex harmonic function, apply the argument to real and imaginary parts. Every live state has a path into $\mathcal{C}$. Applying the same maximum and minimum argument to all states shows that a real harmonic function cannot exceed or fall below the constant already forced on $\mathcal{C}$. Thus y is constant everywhere and x is a multiple of v .

Finally, (33) and the simple factor of p at β imply that no Jordan block at β has size greater than one. Together with the one-dimensional eigenspace, this proves algebraic multiplicity one. $\square$

Proposition 4.5 (Uniform suffix estimate). *In any fixed matrix norm,*

$$A^k = \beta^k \frac{v \ell}{\ell v} + O((7/5)^k). \quad (35)$$

For every $j \in \{3, 4, 5\}$, uniformly over all live states q ,

$$e_q A^k (h_j - d_j \mathbf{1}) = O((7/5)^k). \quad (36)$$

Proof. Lemma 4.4 and (28) identify the spectral projector at β as $v\ell/(\ell v)$; the denominator is nonzero by (31). All remaining eigenvalues lie strictly inside the circle of radius $7/5$, by Lemmas 4.2 and 4.3. In Jordan form their powers are a finite sum of terms bounded by a polynomial in k times $|\lambda|^k$. For each such $|\lambda| < 7/5$, this is $O((7/5)^k)$. Nilpotent blocks vanish after a finite number of steps. This proves (35).

Since $\ell \mathbf{1} = 1$ and $\ell h_j = d_j$, the dominant rank-one term annihilates $h_j - d_j \mathbf{1}$. This gives (36). A single constant works for all states because Q is finite. $\square$

There is no assumption here that the automaton graph itself is one irreducible component. The transient states have been included throughout, which is essential when a counting block starts near the initial state.

Corollary 4.6. *With $c_\beta = (3\beta^2 + 7\beta + 2)/22$, we have*

$$T_k = c_\beta \beta^k + O((7/5)^k), \quad C_j(T_k) = d_j c_\beta \beta^k + O((7/5)^k).$$

Also $d_j > 0$ for $j = 3, 4, 5$ and $d_3 + d_4 + d_5 = 1$.

Proof. Apply (35) to (24), using $v_0 = 1$ and (31). The sum of the three displayed formulas for d_j is exactly one. Positivity can be seen without relying on decimal approximations: the matrices $\beta^{-k} A^k$ are nonnegative and converge to $v\ell/(\ell v)$. Since v and ℓv are positive, ℓ is nonnegative. Its sum is one, and it is supported on $\mathcal{C}$. The left eigenvector equation and strong connectivity force it to be strictly positive at every state of $\mathcal{C}$. Each of the three output classes meets $\mathcal{C}$, so every ℓh_j is positive. $\square$

5 From numeration cutoffs to every integer cutoff

The passage in this section is what turns asymptotic path counts into the natural-density theorem. It also explains why a numerical fit along T_k would have been insufficient.

5.1 The lexicographic decomposition

Take $N \geq 1$ and write its canonical representation as a length- L word

$$b = b_{L-1} \cdots b_0.$$

Pad every representation of an integer below N with leading zeros to length L . For each position at which $b_k = 1$, consider the set of words which agree with b at all more significant positions, have 0 at position k , and then have any allowed suffix of length k . Call this set $\mathcal{B}_k$.

Lemma 5.1 (Disjoint prefix cylinders). *The allowed length- L words representing $0 \leq n < N$ are the disjoint union of the $\mathcal{B}_k$ over positions with $b_k = 1$. If q_k is the automaton state after the retained prefix followed by that 0, then*

$$|\mathcal{B}_k| = e_{q_k} A^k \mathbf{1}, \quad \#\{n \in \mathcal{B}_k : a(n) = j\} = e_{q_k} A^k h_j.$$

Proof. Every smaller word has a unique most significant position where it first differs from b . At that position b has 1 and the smaller word has 0. Conversely, any word formed in this way is lexicographically smaller, and Lemma 2.1 says it is numerically smaller. The initial prefix is allowed because it is a prefix of b with a 1 replaced at its end by 0; that replacement cannot create 111. Legal completions are precisely the suffix paths counted by A^k . Uniqueness of the first position proves disjointness. $\square$

Proof of Theorem 1.1. Subtract d_j times the first count of Lemma 5.1 from the second and sum over the disjoint cylinders. Proposition 4.5 gives

$$\begin{aligned} |C_j(N) - d_j N| &\leq K \sum_{\substack{0 \leq k < L \\ b_k = 1}} (7/5)^k \\ &\leq K \sum_{k=0}^{L-1} (7/5)^k = O((7/5)^L), \end{aligned}$$

where K is independent of the prefix state and of N . There is at most one block of each suffix length, so the geometric sum does not introduce an extra logarithmic factor.

By Lemma 2.1, $T_{L-1} \leq N < T_L$. Corollary 4.6 implies $\beta^L \asymp N$. With θ from (8),

$$(7/5)^L = (\beta^L)^\theta = O(N^\theta).$$

This proves (9). Since $\theta < 1$, division by N proves the ordinary natural-density limits. Positivity and normalization were proved in Corollary 4.6. $\square$

5.2 Two uniform consequences

Corollary 5.2 (Fixed-prefix proportions). *Fix any allowed prefix w , with leading zeros permitted. Among all allowed words ws with $|s| = k$, the proportion whose represented length has additive complexity j tends to d_j as $k \rightarrow \infty$.*

Proof. If q is the state reached by w , the numerator is $e_q A^k h_j$ and the denominator is $e_q A^k \mathbf{1}$. Their leading terms from (35) are, respectively, $\beta^k v_q d_j / (\ell v)$ and $\beta^k v_q / (\ell v)$. The latter is positive and the error is of smaller exponential order. $\square$

Corollary 5.3 (Intervals and the mean complexity). *For integers $X \geq 1$ and $1 \leq H \leq X$,*

$$\#\{X \leq n < X + H : a(n) = j\} = d_j H + O(X^\theta).$$

In particular the corresponding proportions tend to d_j whenever $H/X^\theta \rightarrow \infty$. Furthermore,

$$\frac{1}{N} \sum_{0 \leq n < N} a(n) = \frac{13\beta^2 + 4\beta + 33}{22} + O(N^{\theta-1}), \quad (37)$$

whose limiting value is 3.83344690925626054335...

Proof. Subtract (9) at X from the same formula at $X + H$; because $H \leq X$, both error terms are $O(X^\theta)$. For the mean, the length-zero contribution is 1, and every positive length contributes one of 3, 4, 5. Thus

$$\sum_{n < N} a(n) = 1 + 3C_3(N) + 4C_4(N) + 5C_5(N)$$

for $N \geq 1$. Substitute the three count estimates and simplify $3d_3 + 4d_4 + 5d_5$ in the basis $1, \beta, \beta^2$. $\square$

Remark 5.4 (What the exponent does not say). The exponent in (8) is a certified convenient bound, not an assertion of optimality. It comes from the rational spectral radius bound $7/5$, whose proof is the single exact inequality (34). The theorem supplies no claimed numerical value for the implicit constant in the error term, and it does not assert uniform relative accuracy on intervals shorter than the scale covered by Corollary 5.3.

6 Exact counting and rational generating functions

6.1 An exact algorithm for any cutoff

The proof gives a practical counting algorithm. Let

$$F_k(q, j) = e_q A^k h_j.$$

Its initialization and recurrence are

$$F_0(q, j) = \mathbf{1}_{o(q)=j}, \quad F_{k+1}(q, j) = \sum_{\substack{d \in \{0,1\} \\ \delta(q,d) \text{ legal}}} F_k(\delta(q, d), j). \quad (38)$$

Precompute these arrays for $0 \leq k \leq L$, read the digits of N , and at each digit 1 add the array for the alternative zero-transition and the remaining suffix length. This is exactly the disjoint union used in Lemma 5.1. Only additions of nonnegative integers are required; no density approximation enters the computation.

There are 76 states and four output labels including the empty-factor value 1. The precomputation uses $O(76L)$ integer additions up to a fixed constant, and the final digit scan uses $O(L)$ more. The integers have $O(L)$ bits. With ordinary addition this gives a conservative $O(76L^2)$ bit-operation bound, where $L = O(\log N)$; retaining all suffix layers uses $O(76L^2)$ bits. Computing a single $a(n)$ uses one transition per digit after conversion to Tribonacci numeration.

Here are exact counts produced by the included code. The three displayed counts sum to $N - 1$, because the single length 0 has value 1.

N	$C_3(N)$	$C_4(N)$	$C_5(N)$
10^3	321	594	84
10^6	280,918	607,661	111,420
10^9	279,580,105	607,507,914	112,911,980
10^{12}	279,529,114,618	607,499,398,284	112,971,487,097

The data files also include exact counts at 10^{18} , 10^{30} and 10^{100} . These are finite computations of whole intervals, not extrapolations from smaller samples. For example, at $N = 10^{18}$ the counts are

$$\begin{aligned} C_3(10^{18}) &= 279527022667110675, \\ C_4(10^{18}) &= 607499052340780247, \\ C_5(10^{18}) &= 112973924992109077. \end{aligned}$$

The comparatively slow visible convergence at moderate cutoffs is compatible with the error bound; it was not used to guess exact coefficients in the density formulas.

6.2 Generating functions at Tribonacci cutoffs

The whole sequence of cutoff counts has a compact rational description. Set

$$G_j(z) = \sum_{k \geq 0} C_j(T_k) z^k \quad (j = 3, 4, 5) \quad (39)$$

and define

$$\begin{aligned} B(z) &= 1 - z - z^2 - z^3, \\ Q(z) &= 1 - 2z^3 - 2z^7 - 2z^{11} + z^{12}, \\ D(z) &= (1 - z)^2 B(z) Q(z). \end{aligned} \quad (40)$$

Proposition 6.1 (Exact rational cutoff series). *For $j \in \{3, 4, 5\}$,*

$$G_j(z) = \frac{P_j(z)}{D(z)}, \quad (41)$$

where the integer coefficients of P_j are given in Appendix C. Their degrees are 27, 30, 30, respectively. In addition,

$$\frac{1}{1-z} + G_3(z) + G_4(z) + G_5(z) = \frac{1+z+z^2}{B(z)}. \quad (42)$$

The common denominator displayed here is sufficient for the assertions; no minimality claim is needed.

Finite-certificate proof. For a proposed numerator P_j , compare its coefficients with those of $D(z) \sum_{k \geq 0} e_0 A^k h_j z^k$. The verifier checks the coefficient equalities through degree 106. The degree of D is 17 and the largest numerator degree is 30.

For $k \geq 17$, the residual coefficient is a fixed linear combination of $e_0 A^{k-i} h_j$ for $0 \leq i \leq 17$. Thus the residual sequence, after this initial offset, satisfies the monic order-76 recurrence supplied by $p(A) = 0$. Its 76 consecutive values at $k = 31, \dots, 106$ are all zero. The recurrence forces every subsequent value to be zero. The earlier coefficients have been checked individually, proving (41) identically, rather than to a finite truncation only.

The output 1 occurs exactly for the zero representation, contributing 1 at every cutoff T_k . Summing all four output classes gives T_k . Equation (14) proves (42). $\square$

The dominant-pole ratios provide an additional independent check on the *evaluation* of the three densities. Put $\rho = \beta^{-1}$ and write $d_j = (a_j \beta^2 + b_j \beta + c_j)/22$, with

$$(a_3, b_3, c_3) = (21, -38, 5), \quad (a_4, b_4, c_4) = (-55, 72, 67), \quad (a_5, b_5, c_5) = (34, -34, -50).$$

The verifier checks the exact polynomial divisibilities

$$B(z) \mid 22z^2 P_j(z) - (a_j + b_j z + c_j z^2)(1-z)^2 Q(z)(1+z+z^2). \quad (43)$$

Evaluating at ρ says precisely that the ratio of the dominant pole coefficients in G_j and the total cutoff generating function is d_j . This agrees with the eigenvector aggregation. It is not used as a substitute for the arbitrary-cutoff argument in Section 5.

7 Verification, reproducibility and limits of the claim

7.1 Finite proofs versus finite experiments

The verifier contains two logically different types of check. First, it verifies finite identities that support an all-index proof: closure under both co-decomposition transitions; the full quotient diagram; the right and left eigenvector equations; the integer annihilator; the spectral separation inequality; and the generating-function recurrences. Once verified, these combine with the induction and asymptotic arguments in the article to establish claims for all lengths and all cutoffs.

Second, it performs additional finite cross-checks. These are useful for detecting implementation mistakes but are not, by themselves, evidence that replaces an all-index proof. The most substantial of these is a separate enumeration of *all* length- n factors for every $1 \leq n \leq 2048$.

Lemma 7.1 (Complete finite coverage for the cross-check). *The legal adjacent letter pairs of $\mathbf{t}$ are exactly*

$$\mathcal{P} = \{00, 01, 02, 10, 20\}.$$

If m is chosen so that $\min_a |\varphi^m(a)| \geq H$, then every factor of $\mathbf{t}$ of length at most H occurs in at least one word $\varphi^m(ab)$ with $ab \in \mathcal{P}$. Conversely, every factor of one of those five words occurs in $\mathbf{t}$.

Proof. All five listed pairs occur in $\varphi^3(0) = 0102010$. Inside a single substitution image, the only pairs are 01 and 02. Across two images, the next image begins with 0 and the previous one ends in 0, 1, or 2, so the only possibilities are 00, 10, 20. This proves the exact pair set.

Since $\mathbf{t} = \varphi^m(\mathbf{t})$, the fixed point is a concatenation of blocks $\varphi^m(a)$. A factor of length at most H starting inside one block cannot extend beyond the end of the next block: that next block already has length at least H . Therefore the factor is contained in an image of a legal adjacent pair. Conversely, each legal adjacent pair occurs in $\mathbf{t}$, so its image occurs in $\varphi^m(\mathbf{t}) = \mathbf{t}$. $\square$

For $H = 2048$, the verifier uses $m = 14$, for which the shortest image has length 3136. It forms prefix-sum arrays for the five pair images, slides a window of each length through each image, and collects the exact sums. This examines 93,070,336 windows in total. The resulting number of sums agrees with the automaton at every tested length. Unlike testing a long guessed prefix, this enumeration has a proved coverage guarantee.

7.2 Recorded verification results

The standard run recorded in `data/verification_results.json` returns PASS. Its principal checks are:

Check	Size or outcome
Co-decomposition pairs / closed sets	56 / 277
Reachable valid-language product states	296
Live quotient states / closed component	76 / 66
Left-eigenvector integer component equalities	228
Entries verified zero in $p(A)$	5776
Independent complete factor-length checks	2048
Windows examined in the all-factor check	93,070,336
Individually evaluated lengths for count checks	100,000
Arbitrary prefix-count comparisons	301
Tribonacci cutoff comparisons	61
Generating-function coefficients through degree	106
Independent density-residue polynomial identities	3

The matrix and algebraic checks use exact integers or rational numbers. Decimal expansions are generated only for presentation and do not feed back into a certificate.

7.3 How to reproduce the package

From the extracted package directory, run:

```
python3 code/verify.py
python3 code/tribonacci.py --value 123456789
python3 code/tribonacci.py --count 1000000000000000000
```

The verifier needs no external Python package, no network access, and no Walnut installation. It rebuilds the automaton and compares it with the stored certificate instead of simply trusting the

serialized table. Its checks use explicit exceptions, not Python assertions, so they remain active when invoked with `python3 -O`.

The optional script `code/derive_spectrum.py` uses SymPy to reproduce the symbolic derivation of the annihilator, the left eigenvector and the rational generating functions. It is a discovery/reproduction aid, not a dependency of the standard-library verification. It was tested with SymPy 1.14.0. The script `code/make_tables.py` regenerates the printed table fragments and the exact-count data. The article can be built with

```
pdflatex -halt-on-error article.tex
pdflatex -halt-on-error article.tex
```

A `Makefile` provides the same operations. The original random draw is preserved rather than repeated during verification; its list digest and selected index are checked for consistency.

7.4 Scope, trust and remaining questions

The claim is a complete proposed resolution of the stated natural-density question, supported by an infinite argument and finite exact certificates. It is not an independently refereed theorem, a Lean formalization, or a proof that no previous resolution exists. The trusted computational layer includes the Python interpreter and the executed verifier. The finite tables and source code are provided so that another implementation can check the same identities independently.

Several broader questions are not settled here: the optimal discrepancy exponent; an optimal explicit error constant; the corresponding density spectrum for arbitrary letter weights; and general density theorems for all substitutive additive-complexity sequences. In particular, the general regularity questions discussed in [1, 3] are not being relabelled as consequences of this specific computation.

8 Conclusion

The additive-complexity proportions of the Tribonacci word are governed by an algebraic left eigenvector of a finite transition matrix. The exact weights are not uniform over recurrent states, but their output-class sums collapse to three small expressions in the Tribonacci constant. A spectral gap holds uniformly from every live state, and the lexicographic prefix-cylinder decomposition transfers that estimate to every integer cutoff. This proves the proposed densities and the power-saving estimate, while the archived word-level and integer-matrix certificates make the finite part of the argument reproducible.

A The complete 76-state transition table

The initial state is 0. Read digits most significant first. The column o is the output, and a dash is an illegal transition to the omitted sink. State 0 has output 1 and loops on 0, accounting for leading zeros. The two halves of the table together list every state exactly once.

q	$\delta(q, 0)$	$\delta(q, 1)$	o	q	$\delta(q, 0)$	$\delta(q, 1)$	o
0	0	1	1	38	46	14	4
1	2	3	3	39	29	47	3
2	4	5	3	40	48	49	5
3	6	–	3	41	50	–	4
4	7	8	3	42	51	18	5
5	9	10	4	43	28	52	4
6	11	5	4	44	53	26	5
7	6	12	3	45	46	31	4
8	13	14	3	46	32	19	5
9	15	5	4	47	35	–	4
10	16	–	3	48	21	54	3
11	17	18	4	49	30	10	5
12	6	3	3	50	33	55	4
13	4	19	3	51	56	23	5
14	20	–	3	52	57	–	3
15	21	22	3	53	58	38	5
16	11	19	4	54	59	60	4
17	20	23	3	55	40	61	4
18	16	14	3	56	62	45	4
19	9	24	4	57	63	64	4
20	25	26	4	58	65	45	4
21	27	18	4	59	48	66	4
22	28	14	4	60	67	–	4
23	27	10	4	61	68	–	4
24	29	–	3	62	62	45	5
25	30	18	5	63	30	39	5
26	28	31	4	64	69	70	4
27	32	5	5	65	71	26	5
28	15	19	4	66	30	24	5
29	33	34	4	67	25	72	4
30	35	23	4	68	33	64	4
31	36	–	3	69	48	66	5
32	37	38	4	70	57	–	4
33	17	39	4	71	62	38	5
34	40	41	4	72	59	73	4
35	42	26	4	73	74	–	4
36	25	43	4	74	25	75	4
37	44	45	4	75	59	70	4

B The exact normalized left eigenvector

The table gives the nonzero entries of $\ell_q = (u_q + w_q\beta + z_q\beta^2)/44$. The entries at 0, 1, 2, 3, 4, 6, 7, 8, 12, 13 are zero. All coefficients are integers; (29) is sufficient to verify $\ell A = \beta\ell$.

q	u_q	w_q	z_q	q	u_q	w_q	z_q
5	-22	-204	118	43	-6	-26	16
9	-32	118	-54	44	-6	-26	16
10	81	242	-155	45	-124	-142	114
11	-100	-48	56	46	-150	12	38
14	-185	122	-11	47	228	-168	24
15	56	-44	8	48	-13	-2	5
16	56	-44	8	49	-190	-18	66
17	161	-236	81	50	-55	-16	25
18	-73	-178	119	51	-20	22	-6
19	-32	118	-54	52	-20	22	-6
20	-90	94	-24	53	-20	22	-6
21	-89	-30	43	54	11	18	-13
22	-100	-48	56	55	39	80	-55
23	57	148	-97	56	42	14	-20
24	-11	-260	145	57	319	-6	-91
25	314	150	-174	58	42	14	-20
26	124	132	-108	59	-19	-260	147
27	150	-22	-32	60	7	-24	11
28	40	-54	18	61	41	-94	39
29	-21	-12	13	62	14	-48	22
30	-238	174	-24	63	-325	-410	319
31	-10	6	0	64	-192	-234	184
32	-10	6	0	65	-28	-62	42
33	181	290	-211	66	177	16	-61
34	9	34	-21	67	-31	4	7
35	16	10	-10	68	-135	-2	41
36	16	10	-10	69	-42	376	-192
37	16	10	-10	70	-71	206	-91
38	162	30	-64	71	-34	70	-28
39	24	252	-144	72	35	38	-31
40	66	-124	48	73	3	-66	35
41	25	-30	9	74	-69	32	3
42	-6	-26	16	75	101	72	-69

The coefficient sums are (44, 0, 0), so $\ell\mathbf{1} = 1$ exactly. Summing only rows with a specified output gives the three density formulas in Theorem 1.1. The machine-readable version is `data/eigenvector.json`.

C Numerators of the rational cutoff generating functions

For $j = 3, 4, 5$, write $P_j(z) = \sum_{k=0}^{30} p_{j,k} z^k$ and use the common denominator $D(z)$ of (40). Coefficients beyond the displayed range are zero. This table, together with D , fully specifies Proposition 6.1.

k	$p_{3,k}$	$p_{4,k}$	$p_{5,k}$
0	0	0	0
1	1	0	0
2	0	0	0
3	-3	2	0
4	-1	-1	0
5	2	-2	0
6	-1	3	0
7	3	-7	4
8	-4	8	-6
9	1	-3	2
10	7	-13	8
11	-7	14	-7
12	1	3	-6
13	11	-25	15
14	-10	20	-8
15	3	-3	-1
16	5	-9	4
17	-6	8	-2
18	1	-1	0
19	3	-9	6
20	-4	17	-13
21	4	-13	9
22	-2	0	2
23	0	4	-4
24	-1	4	-3
25	1	-8	7
26	1	2	-3
27	-1	1	0
28	0	2	-2
29	0	-3	3
30	0	1	-1

D Artifact map and selection record

The core package has the following organization. The README gives individual filenames and command-line examples for every script.

Path	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Article source and typeset report.
<code>code/tribonacci.py</code>	Exact reconstruction, single-value evaluation and prefix counting.
<code>code/verify.py</code>	Standard-library verifier of the finite certificates and cross-checks.
Other scripts in <code>code/</code>	Optional symbolic derivation, table regeneration and random-selection auditing.
Automaton data in <code>data/</code>	All 277 co-decomposition sets, the 76-state machine and its quotient information.
Algebraic data in <code>data/</code>	Exact eigenvector triples, generating functions and decimal constants.
Count data in <code>data/</code>	Exact counts at large cutoffs, including 10^{100} , in JSON and CSV form.
Verification logs in <code>data/</code> <code>data/selection.json</code> <code>notes/</code>	Recorded outcomes, bounds and actual verification scope. Original ordered 120-area list and the single recorded selection. Source-version and manifest-exclusion audits, plus the original area-draw script.
<code>STATUS.md</code> , <code>README.md</code>	Claim boundaries and practical reproduction instructions.

The original draw selected item 68, *Combinatorics on words*. The SHA-256 of the ordered list, encoded as UTF-8 strings separated by a single newline, is

ea66507989230ed06b68907eef7e2e8c14f
933ba70aa97b56a70fab0ea4477b6.

The two lines above are one concatenated digest, with no intervening space. The digest authenticates the recorded list against accidental alteration; it is not an external timestamp or a cryptographic proof of the historical random-draw procedure. The record reports the actual executed single draw.

References

- [1] Pierre Popoli, Jeffrey Shallit and Manon Stipulanti, *Additive Word Complexity and Walnut*. In *44th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2024)*, LIPIcs **323** (2024), Article 32, 32:1–32:18. [doi:10.4230/LIPIcs.FSTTCS.2024.32](https://doi.org/10.4230/LIPIcs.FSTTCS.2024.32). Full preprint: [arXiv:2410.02409v1](https://arxiv.org/abs/2410.02409v1), 3 October 2024. The theorem and remark numbers in this article refer to Theorem 18 and Remark 19 of that full preprint.
- [2] Ondřej Turek, *Abelian Complexity Function of the Tribonacci Word*. Journal of Integer Sequences **18** (2015), Article 15.3.4. <https://cs.uwaterloo.ca/journals/JIS/VOL18/Turek/turek3.html>. Also [arXiv:1309.4810v2](https://arxiv.org/abs/1309.4810v2). The co-decomposition approach is prior work; its specialization needed here is proved in Section 3.
- [3] Jean-Michel Couvreur, Martin Delacourt, Nicolas Ollinger, Pierre Popoli, Jeffrey Shallit and Manon Stipulanti, *Effective Computation of Generalized Abelian Complexity for Pisot Type Substitutive Sequences*. [arXiv:2504.13584v2](https://arxiv.org/abs/2504.13584v2), 22 April 2025. Consulted for the current literature audit and related automata constructions; not imported as a density theorem.